

Academic Research Digest — Learning Environments

Window covered: 22 August – 21 September 2026 (30 days) · **Compiled:** 21 September 2026

Note on scope: the standing brief asks for the "last 7 days" in its framing but sets a 30-day cutoff in the selection criteria. As in the previous digest, this edition uses the 30-day window so genuinely new items aren't missed, and states each paper's exact date so the reader can see how recent it is within that window. All four papers below cluster in the 27–28 August window; no items dated after that were found meeting both the date and keyword criteria (see transparency note).

AI & Learning Environments

Reclaiming Epistemic Agency: A Critical Framework for Human-Generative AI Co-Agency in Education

- **Authors:** Biranchi Poudyal
- **Source:** arXiv preprint (arXiv:2608.26937)
- **Publication date:** Submitted 27 August 2026
- **URL:** <https://arxiv.org/abs/2608.26937>

This conceptual paper argues that generative AI does more than assist instruction — it redistributes *epistemic authority* (the power to produce knowledge and validate claims) among teachers, students, and machines, drawing on Distributed Agency, Self-Determination Theory, Society 5.0, and the TPACK/SAMR technology-integration frameworks. For teaching quality, it reframes GenAI-integration choices (where a tool sits on the SAMR ladder, say) as decisions about *who* gets to judge student work, not just which tool to adopt. Through the SDT lens, autonomy becomes contested terrain: GenAI can expand students' sense of self-directed inquiry while simultaneously narrowing the competence-relevant feedback loops that come from teacher and peer validation, if machine judgment displaces them.

Comparative Learning Environments Research

What University signs might tell us about contemporary pedagogic practices

- **Authors:** Emily Danvers, Susanne Strömberg Jämsvi
- **Source:** *Learning Environments Research* (Springer), Vol. 29, Article 48
- **Publication date:** Published online 28 August 2026
- **URL:** <https://link.springer.com/article/10.1007/s10984-026-09601-3>

Comparing signage across teaching and learning spaces at an English and a Swedish university through photomapping and critical semiotic analysis, the study finds classroom signs frequently construct "impersonal and generic" spaces that privilege technological systems and frame embodied student presence as a potential disruption. For instructional effectiveness, this surfaces an inexpensive, overlooked

lever — the everyday material environment — that can invite or dampen active, embodied participation independent of formal curriculum design (the authors note changing signage is cheaper than rebuilding a room). Through an SDT lens, signage that governs bodies and normalizes passive, device-mediated behavior can quietly undercut autonomy (self-directed use of physical space) and relatedness (visible, embodied interaction), even where the taught curriculum otherwise supports both.

SDT, Belonging & Student Well-Being

The Relational Online Learner: Self-Directed Learning, University Belonging, and Student Engagement in Online Higher Education from a Self-Determination Theory Perspective

- **Authors:** Not reliably identified from available sources (see transparency note)
- **Source:** *Education Sciences* (MDPI), Vol. 16, Issue 9, Article 1355
- **Publication date:** September 2026 (issue date; exact publication day not confirmed — see transparency note)
- **URL:** <https://doi.org/10.3390/educsci16091355>

Surveying 696 students at a large open/distance-learning university in Türkiye, the study finds university belonging significantly predicts every measured dimension of student engagement, and is itself shaped by self-control, learning skills, and sustained desire to learn — but not by metacognitive awareness or source-identification ability. For teaching quality, this suggests online course design should treat belonging-building as a lever for engagement broadly rather than a soft add-on, while cautioning that generic "learn how to learn" skills training may not be the mechanism that actually builds it. Through SDT, the results position belonging (relatedness) as a driver that feeds into engagement partly via autonomy-supportive self-directed-learning capacities, reinforcing relatedness and autonomy as interlocking rather than independent needs.

Faculty leadership and psychological safety: a qualitative evaluation of a graduate gross anatomy course redesign

- **Authors:** Gail Elliott, Justin Kowal, Grace Pinhal-Enfield
- **Source:** *Learning Environments Research* (Springer), Vol. 29, Article 47
- **Publication date:** Published online 27 August 2026
- **URL:** <https://link.springer.com/article/10.1007/s10984-026-09604-0>

Thematically analyzing open-ended course-evaluation comments from 88 graduate students in a redesigned gross anatomy course at Rutgers, using Moos' learning-environments framework, the study finds faculty accessibility, mentorship, and humour were the recurring drivers of students' sense of psychological safety. For teaching quality, this locates psychological safety not in curriculum content but in day-to-day faculty behavior — approachability and relational warmth — as the lever program directors redesigning high-stakes technical courses should prioritize. Through SDT, faculty accessibility and mentorship map directly onto relatedness, and the paper's framing of psychological safety as "fundamental to engagement and learning" ties relatedness support directly to competence development in a historically anxiety-prone course.

Search Scope & Transparency Note

Databases/sources queried: Live web search only (Google Scholar results via general web search, arXiv abstract pages, *Learning Environments Research*/Springer, *Education Sciences*/MDPI, Frontiers, Taylor & Francis, ScienceDirect, ResearchGate, Times Higher Education, and general web). Scopus was not separately queried — no unauthenticated Scopus access path exists in this environment, so the search went straight to the specified fallback chain (Google Scholar / web search).

What worked: Web search surfaced candidate papers across all requested theme areas, including two *Learning Environments Research* articles, one arXiv preprint, and one MDPI journal article, all dated within the 30-day window.

What didn't, and why: As in the previous digest, this session's network egress policy blocks direct HTTP(S) access to `arxiv.org` (including its API), `link.springer.com`, `api.semanticscholar.org`, and `api.openalex.org` (confirmed via direct `curl`/fetch attempts returning `403`/`EGRESS\_BLOCKED`/connection-rejected errors). This ruled out:

- Native ArXiv API/`arxiv`-package queries and the `scholarly` package — fell back to web-search snippets of arXiv abstract pages.
- OpenAlex and Semantic Scholar API queries (the protocol's Scopus fallback) — both blocked outright, so Scopus coverage relies entirely on Google Scholar/general web search, which is a weaker substitute.
- Direct verification of full text on Springer/MDPI pages — bibliographic details above come from search-engine snippets and cached summaries, not the primary page. This is a materially weaker source and produced one confirmed gap: the *Education Sciences* article's author list could not be recovered from any snippet and is reported as unresolved rather than guessed.
- ResearchGate/Academia.edu: queried via web search per protocol (direct access would hit login walls); no ResearchGate- or Academia.edu-hosted paper within the 30-day window met the keyword criteria.
- Dr. Phil's Newsletter (Substack): located and checked, but it is a non-peer-reviewed practitioner newsletter — consistent with the brief's "peer-reviewed academic papers" scope, no items from it are included here.

Selection outcome: Several additional candidates were found and excluded, mostly because their true publication/submission dates fell well outside the 30-day window despite surfacing in "2026" searches — a large cluster of AI-in-education and student-agency arXiv preprints (IDs from January through June 2026) and a *Learning Environments Research* article by Burns, Luo & Perlman (published 17 August 2026, four days short of the window) were all excluded on date grounds. One additional occupational-therapy learning-environments study from the prior digest's window was not re-included.

Net result: 4 papers met both the date and keyword criteria with reasonably confirmable bibliographic details, spanning all three requested themes — one more than the prior edition, though still concentrated in the 27–28 August publication window rather than spread evenly across the full 30 days.